Tesselle

YOUR SHOPPING AGENT

GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

One agent for everything that has to fit.

Say it, show it, or look at it through your glasses. Tesselle draws ideas into your own photo, finds real products in stores that deliver to you, calls real makers, and orders - checking every step against your must-haves.



project-tesselle.web.app

Demo · project-tesselle.web.app/demo/tesselle-demo.mp4

github.com/ashfaque-rifaye/project-tesselle

ASHFAQUE RIFAYE · TEAM TESSELLE

01 · THE PROBLEM

Finding things is easy. Getting what fits is not.

≠

Similar isn't suitable

The size of the gap, the bag or the blouse lives in your head, not in the cart. Filters can't check it, so it comes back.



Makers are offline

Tailors, carpenters and print shops answer the phone and WhatsApp, not search boxes. Three quotes take days of calls.



Every device starts over

You research on the phone, compare on the laptop, measure in the room. Nothing carries over, and nothing is checked.

The missing piece isn't more products. It's an agent that **does the whole job** - and holds every store, maker and delivery to what you need.

02 · THE DATA

The cost of guessing - and the shift to agents.

$890B

of goods returned to US retailers in 2024

NRF & HAPPY RETURNS · DEC 2024

~70%

of online clothing returns cite size and fit

CORESIGHT RESEARCH · 2026

25–40%

return rate in online fashion; size & fit is the #1 driver

ZALANDO RESEARCH · KDD 2021

$170–180B

India e-retail by 2030, from $65–66B in 2025 · 290–300M shoppers

BAIN & FLIPKART · HOW INDIA SHOPS ONLINE 2026

7.2 crore

registered MSMEs - the makers custom orders depend on

MINISTRY OF MSME · LOK SABHA, DEC 2025

$3–5T

global commerce AI agents could orchestrate by 2030 · $1T in US retail

MCKINSEY · 2025

So what: returns are a fit problem, India's next 100M shoppers are coming online, and the makers who can make things exactly are still offline. An agent that owns fit end to end sits where these meet.

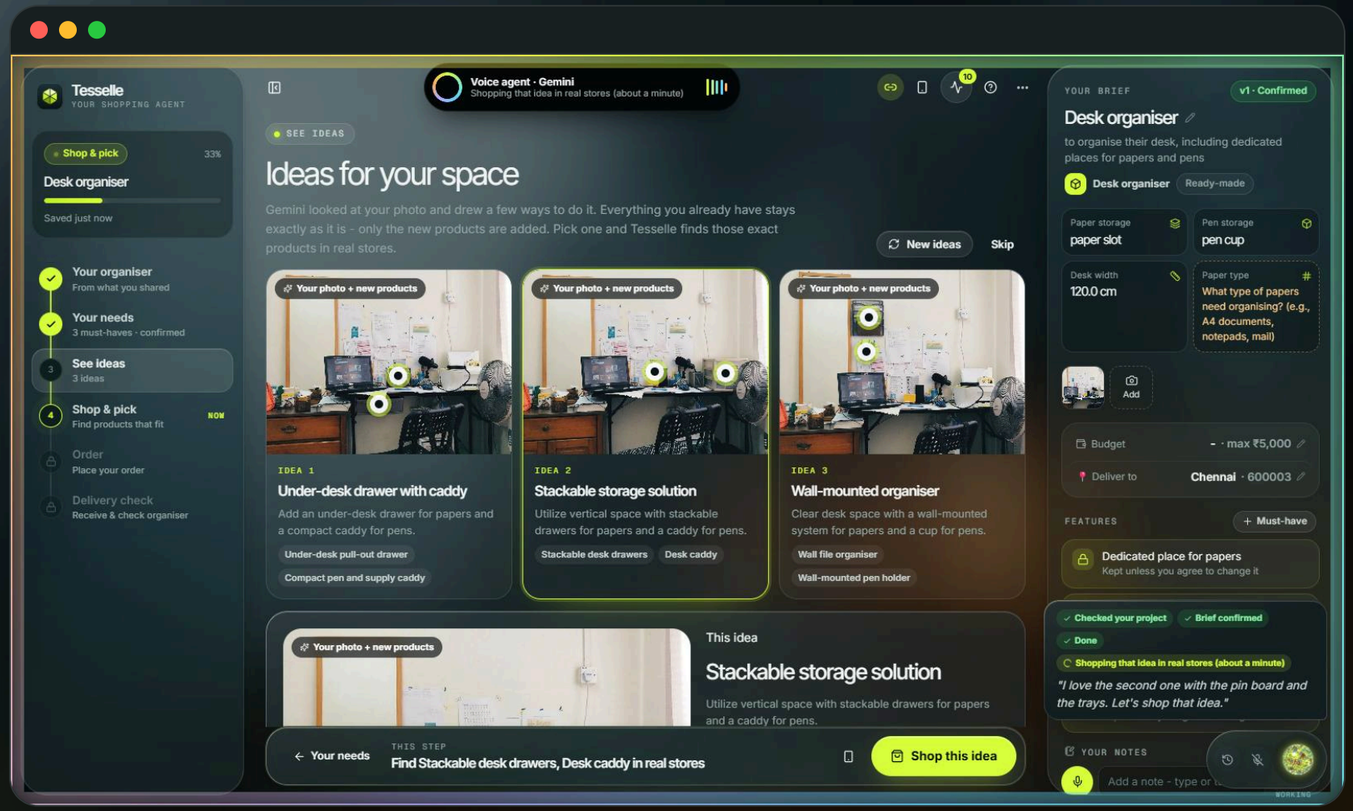
One agent. Real stores, real makers.

Understand - from a sentence or a glance

Voice, text, photos, video or MetaGlass. Gemini writes the brief and the few facts that decide fit for *this* purchase.

See, shop, ask, order

Ideas drawn into your own photo, real products checked against every must-have, makers called and emailed for quotes, and a checkout agent that fills the cart - you place the order.



04 · ADAPTIVE JOURNEYS

Every purchase gets its own steps.

Use case 1 · A messy desk

1. **Your space**
   MetaGlass sees the desk; voice fills the brief
2. **Size & style**
   Fit facts: paper size, pen cup, desk width, ₹5,000
3. **See ideas**
   3 ideas drawn into your own photo
4. **Shop & pick**
   Real products, shop the look
5. **Order**
   Checkout agent · cash on delivery

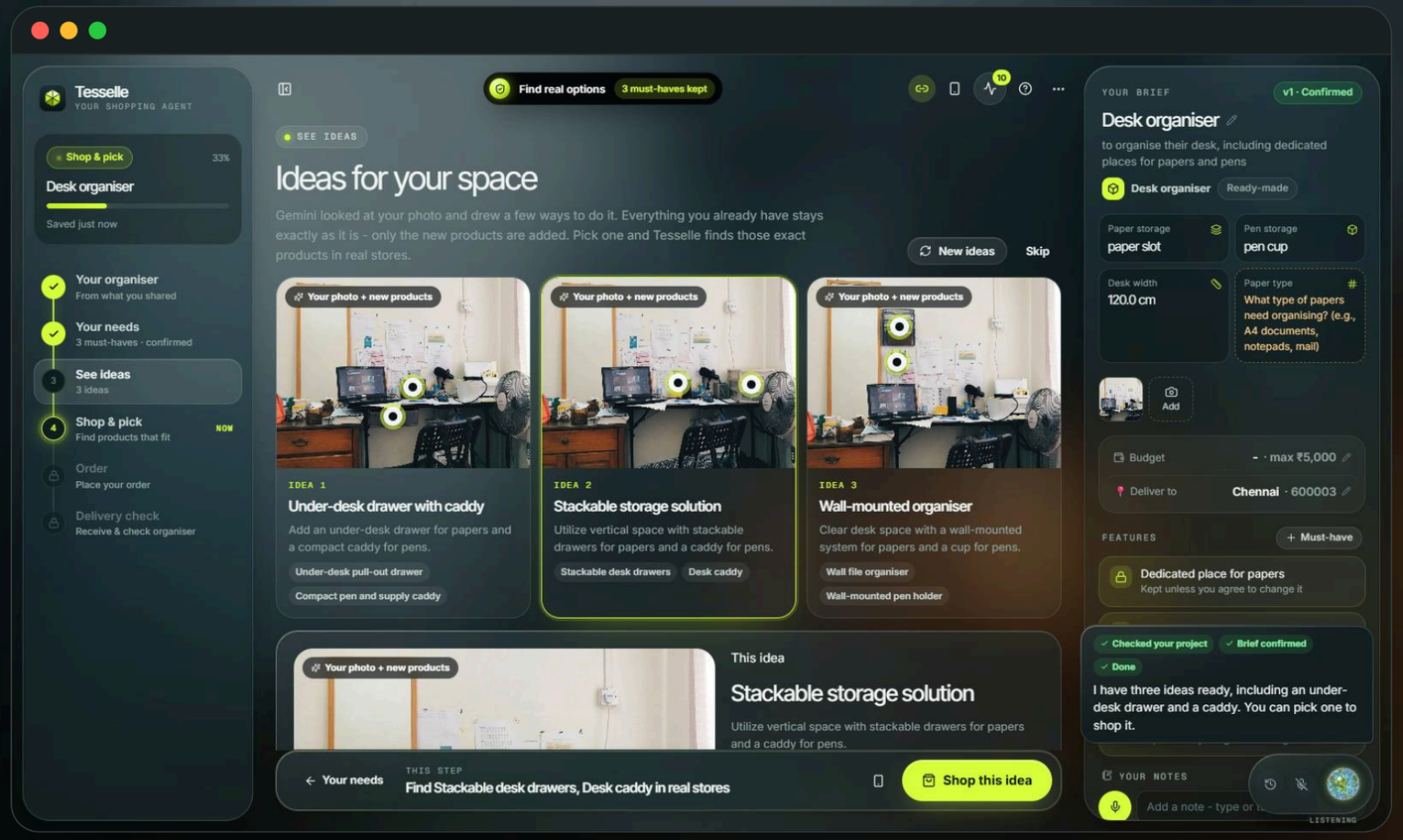
Use case 2 · 500 printed jute tote bags

1. **Specs**
   Quantity, 15×16 in, 2-colour logo, Pune, 3 weeks
2. **Find suppliers**
   Real print shops; contacts from their own sites
3. **Get quotes**
   Email, WhatsApp or a live AI phone call
4. **Compare quotes**
   Each reply checked against the specs
5. **Order & quality check**
   Photo of the delivery vs the brief

Gemini plans and names the steps for each purchase; the fit checker that decides stays deterministic.

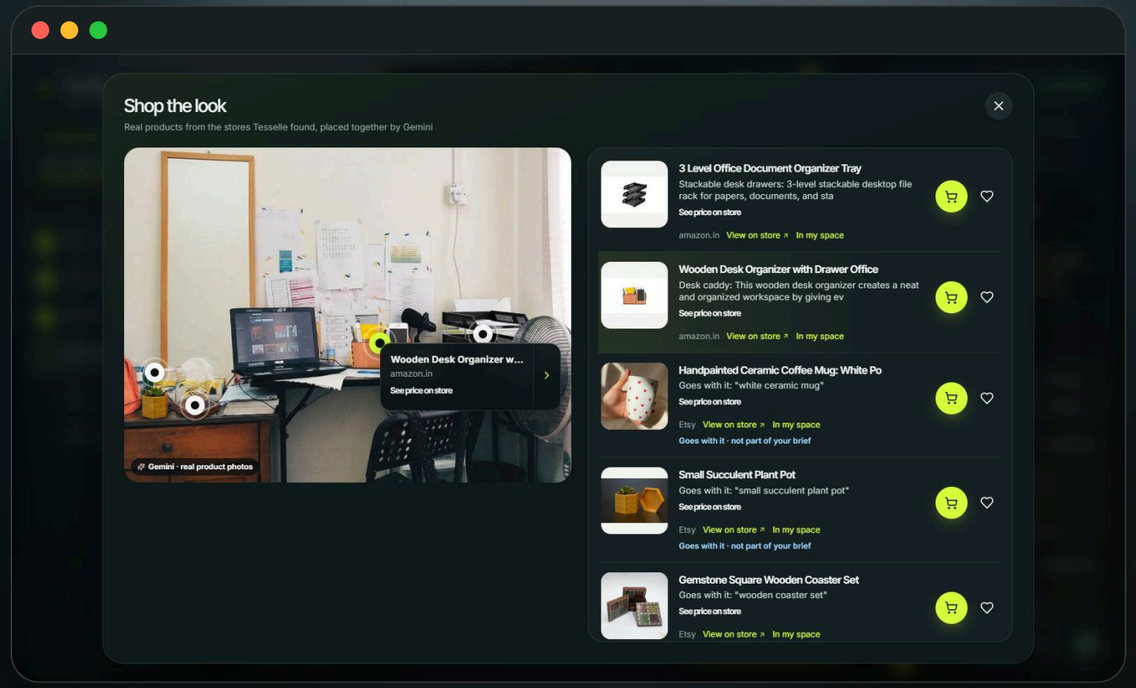
05 · SEE IDEAS

Ideas drawn into your own photo.



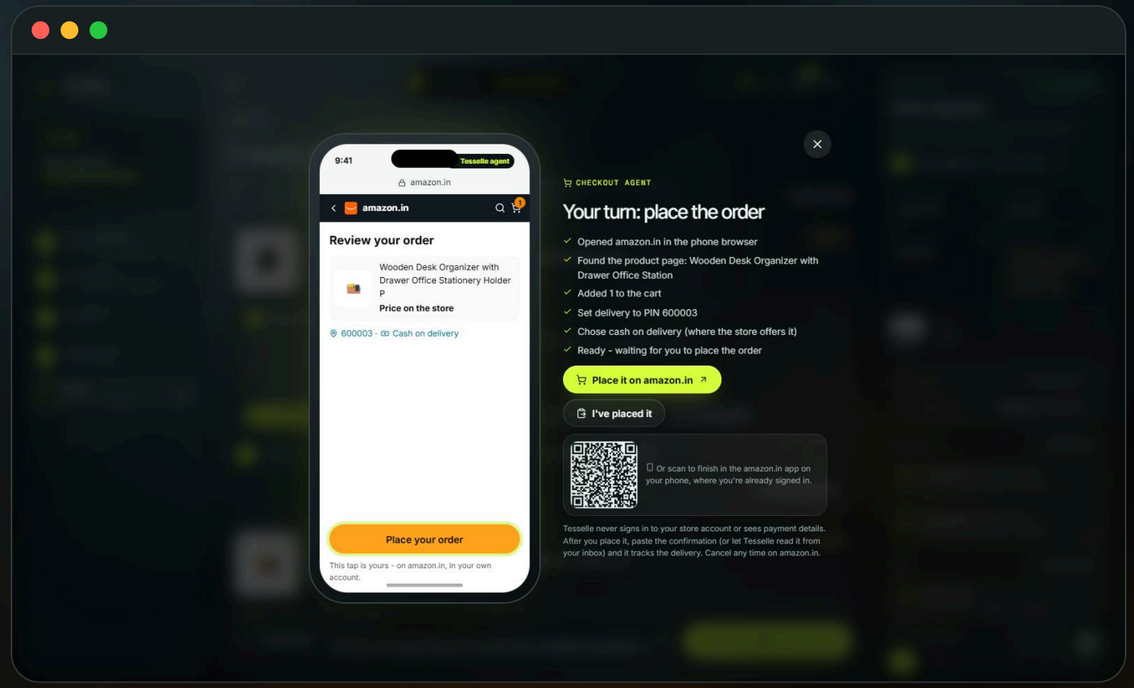
- Gemini looks at the space and draws **2–3 different ideas** - everything you own stays exactly where it is.
- Each idea is **tagged with the real products** it needs.
- "Make it narrower" redraws one; **"shop the second one"** searches real stores for exactly those products.

Real products, a real cart, cash on delivery.



Shop the look

Every dot is a real listing from a store that delivers to your PIN - add to picks or save, like a catalogue page.

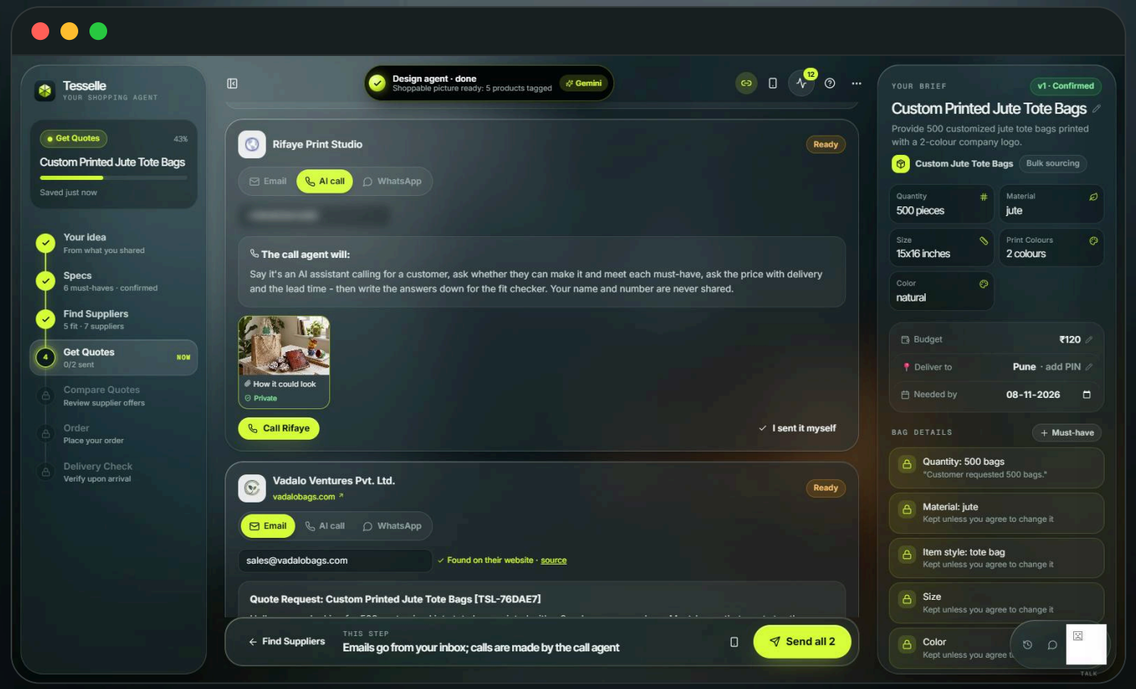


The checkout agent

Walks the store's checkout - product, cart, PIN, cash on delivery - and stops at "Place order". That tap is yours.

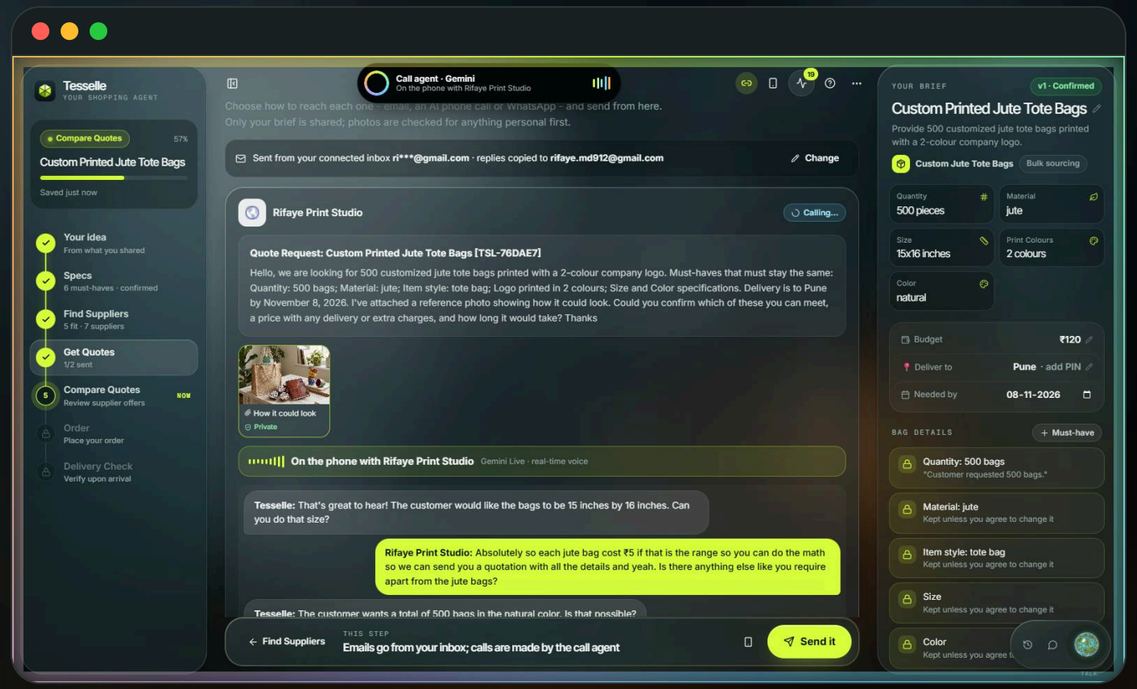
07 · ASK MAKERS

Real makers, reached the way they work.



Contacts found, message written

The outreach agent reads each business's own contact page, then writes the quote request with privacy-checked photos.



A real phone call, in real time

Gemini Live on the phone line says it's an AI, asks size, price and lead time, emails the details after the call, and turns the answers into a checked quote.

08 · GLASSES, LAPTOP, PHONE

From a glance to a doorstep, on any device.

Look at it

MetaGlass: Gemini Live hears you and sees what you see, about a frame a second.

Say it

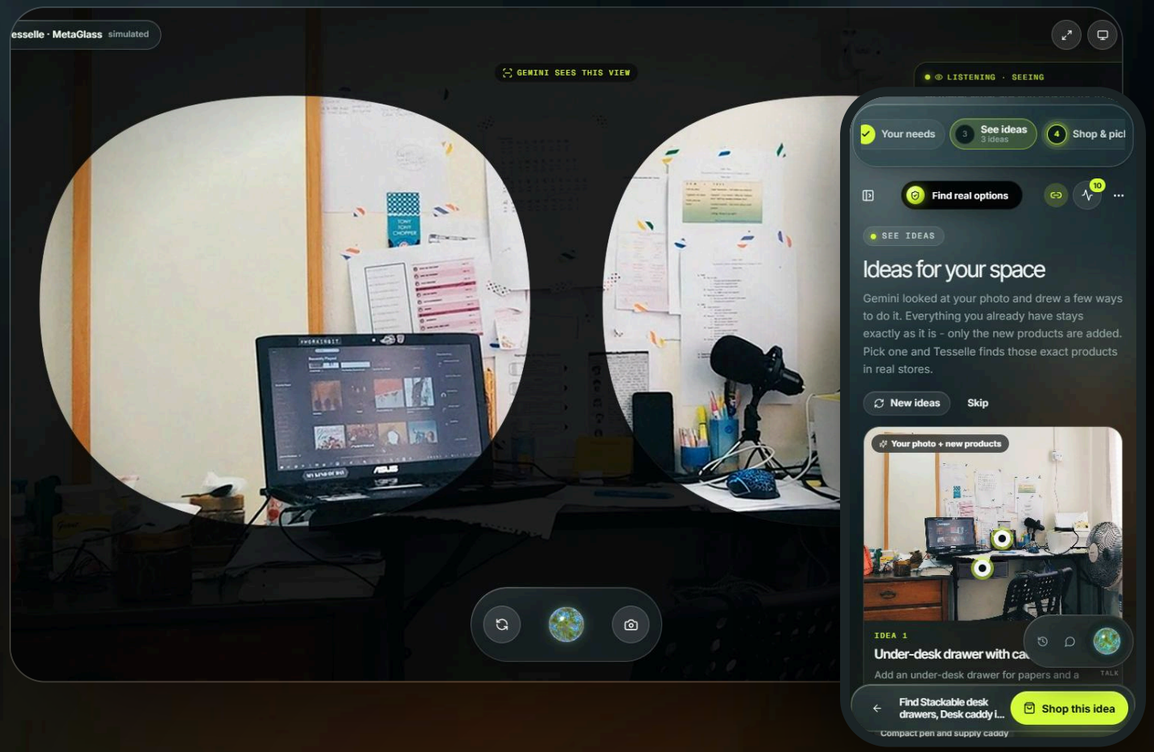
A voice agent that waits for you to finish, then does the work.

Continue anywhere

Laptop and phone mirror each other's step, scroll and agent activity.

Pick up where you left

The conversation resumes exactly where it stopped; replay any turn.

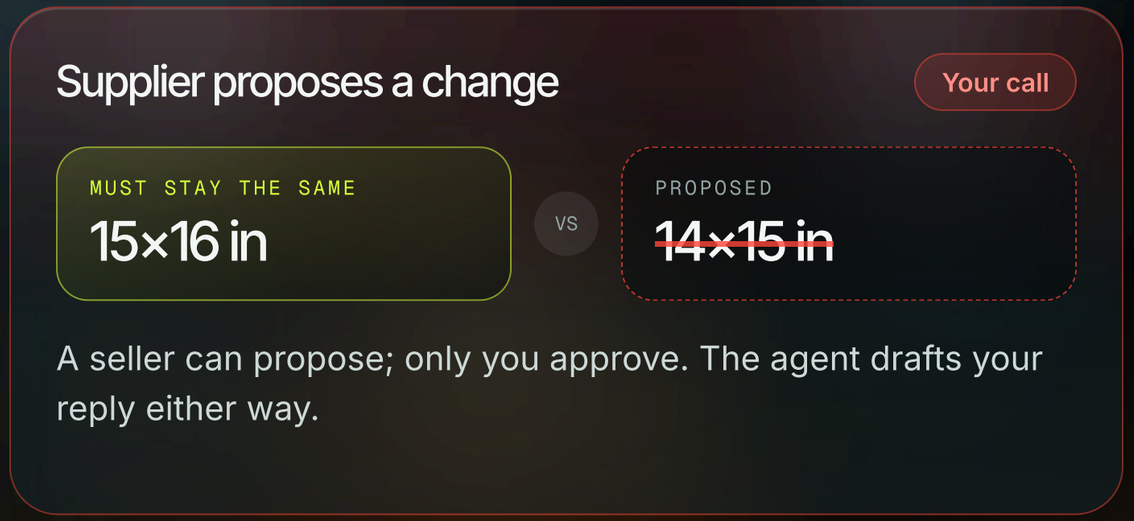


MetaGlass is a simulation of a smart-glasses app; real-glasses capture is on the roadmap.

09 · HOW YOUR MUST-HAVES STAY SAFE

Language models propose. Deterministic code decides.

- **Fit verdicts** - fits / check a detail / doesn't fit, computed in code from structured requirements.
- **Quote deviations** - every offered spec is diffed against the brief version it answers.
- **Versioned brief** - nothing important changes without your decision; restore any version.
- **Honest uncertainty** - photo sizes stay "measure to confirm"; unknown charges never join a total.



Gemini agents that act, grounded in the real web.

VOICE + VISION

Real-time voice agent

Gemini Live native audio drives the app with tools, sees through the camera, resumes sessions and replays turns.

BRIEF + JOURNEY

Brief writer & planner

Typed output: must-haves, fit facts and the steps for this purchase, re-planned as the brief changes.

IDEAS

Design agent

Gemini image editing draws ideas into the customer's photo; products located for tags; shop the look.

SEARCH

Shopping agent + maker scout

ADK graph workflow in parallel: Vertex AI Search over 45 Indian stores + Grounding with Google Search.

OUTREACH

Outreach & call agents

Finds published contacts, writes requests, phones makers over Twilio + Gemini Live, emails the follow-up.

ORDER

Checkout agent & memory

Fills store carts with cash on delivery; Agent Engine Sessions + Memory Bank remember preferences.

Built end-to-end on Google Cloud.

SURFACES

React liquid-glass workspace
Laptop · phone · MetaGlass, live-synced
Voice and camera over WebSockets
Makers: phone (Twilio), WhatsApp, Gmail

Firebase Hosting CDN

CLOUD RUN · FASTAPI + GOOGLE ADK

LlmAgents with typed outputs & tools
Graph workflow (parallel search + join)
Fit checker - deterministic code
Gemini Live bridges mic and phone audio
NDJSON progress streams
One service identity · no API keys

VERTEX AI

Gemini 3 Flash (text, vision, video)
Gemini Live native audio
Gemini image · Gemini TTS
Vertex AI Search (45 stores)
Grounding with Google Search
Agent Engine: Sessions + Memory Bank

FIRESTORE

Projects, brief versions, options, calls, quotes, live device sync

CLOUD STORAGE

Photos, ideas, voice turns, call recordings

SECRET MANAGER

Owner key, Gmail and Twilio credentials

BUILD · LOGGING

Cloud Build, Artifact Registry; every Gemini call labelled

Honest by default. Safe by construction.

- Every option links to a live product page or a real business; prices say "check the store page".
- Must-haves change only through a labelled decision that creates a new brief version.
- Ideas keep everything you own; only new products are added - and labelled as ideas.
- Contacts come only from what businesses publish.

- The agent says it's an AI on every call; recordings only with consent.
- Messages, calls and orders need your yes; the final "Place order" tap is always yours.
- Photos are scrubbed (location data, faces, documents) before a seller sees them.
- Vertex AI only, secrets in Secret Manager, daily cost guard.

From one desk to how India buys anything custom.

NOW

Live agent

Ideas in your photo, real stores and makers, live AI calls, checkout agent, three devices in sync.

NEXT

Makers at scale

WhatsApp Business, Indian phone numbers, Hindi and Tamil voices.

THEN

One-tap orders

Agent-ready checkout with store and UPI partners; real glasses app.

LATER

B2B sourcing

Bulk briefs, samples, pooled demand when minimum quantities block a maker.

Model

Free for shoppers. A disclosed fee on orders and quotes won; subscriptions for makers who want qualified, structured briefs instead of vague enquiries.

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TRY IT · NO SIGN-UP

Everything that has to fit.

project-tesselle.web.app

Demo video · project-tesselle.web.app/demo/tesselle-demo.mp4

Code · github.com/ashfaque-rifaye/project-tesselle



GOOGLE CLOUD AI BUILDER CUP 2026 · THANK YOU